new photon detector next to the beam splitter. If the new detector is switched off (see Figure 7.2b), then we are back in the original experimental setup and the photons generate an interference pattern on the photographic screen. But if the new detector is switched on (Figure 7.2a), it tells us which path each photon traveled: if it detects a photon, then the photon took that path; if it fails to detect a photon, then the photon took the other path. Such “which-path” information, as it’s called, compels the photon to act like a particle, so the wavelike interference pattern is no longer generated.

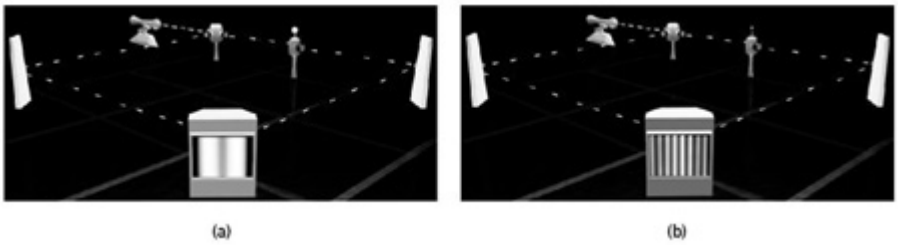


Figure 7.2 (a) By turning on “which-path” detectors, we spoil the interference pattern. (b) When the new detectors are switched off, we’re back in the situation of Figure 7.1 and the interference pattern gets built up.

Now let’s change things, à la Wheeler, by moving the new photon detector far downstream along one of the two pathways. In principle, the pathways can be as long as you like, so the new detector can be a considerable distance away from the beam splitter. Again, if this new photon detector is switched off, we are in the usual situation and the photons fill out an interference pattern on the screen. If it is switched on, it provides which-path information and thus precludes the existence of an interference pattern.

The new weirdness comes from the fact that the which-path measurement takes place long *after* the photon had to “decide” at the beam splitter whether to act as a wave and travel both paths or to act as a particle and travel only one. When the photon is passing through the beam splitter, it can’t “know” whether the new detector is switched on or off—as a matter of fact, the experiment can be arranged so that the on/off switch on the detector is set *after* the photon has passed the splitter. To be prepared for the possibility that the detector is off, the